

X3-Hybrid G4 parallel solution

Parallel solution will offer your PCS greater backup robustness



Storage&Battery

Introduction

The Nominal AC output power of X3-Hybrid G4 is restricted to 5.0-15kW, therefore, it can't be applied to the scenarios where users require more power. Considering this, SolaX provide a solution -- X3-Hybrid parallel solution, which makes it to multiply the power range. In this solution, there are 2 methods to transfer between **On-grid** & **Off-grid** status:

1. **Internal ATS** (Up to **3** inverters parallel; **External ATS** should be set as **Disable**)
2. **External ATS** through **EPS Parallel Box** (Up to **10** inverters parallel, but switching time is longer; **External ATS** should be set as **Enable**)

Notes: Before operation, please make sure that the inverter meets the following four conditions.

1. The **software version** of all inverters shall be the **same**;
2. The **power range** of all inverter models shall be the **same**;
3. The **type and quantity of batteries** connected to all inverters shall be the **same**, or this function cannot be used.
4. **X3 Matebox-advanced** is not compatible to parallel solution.

Advantages

Robust backup capacity

More energy stored will make user deal with Off-grid situation more leisurely.

Flexible

Greater power range can meet more requirements, and be applied to more scenarios.

Configuration

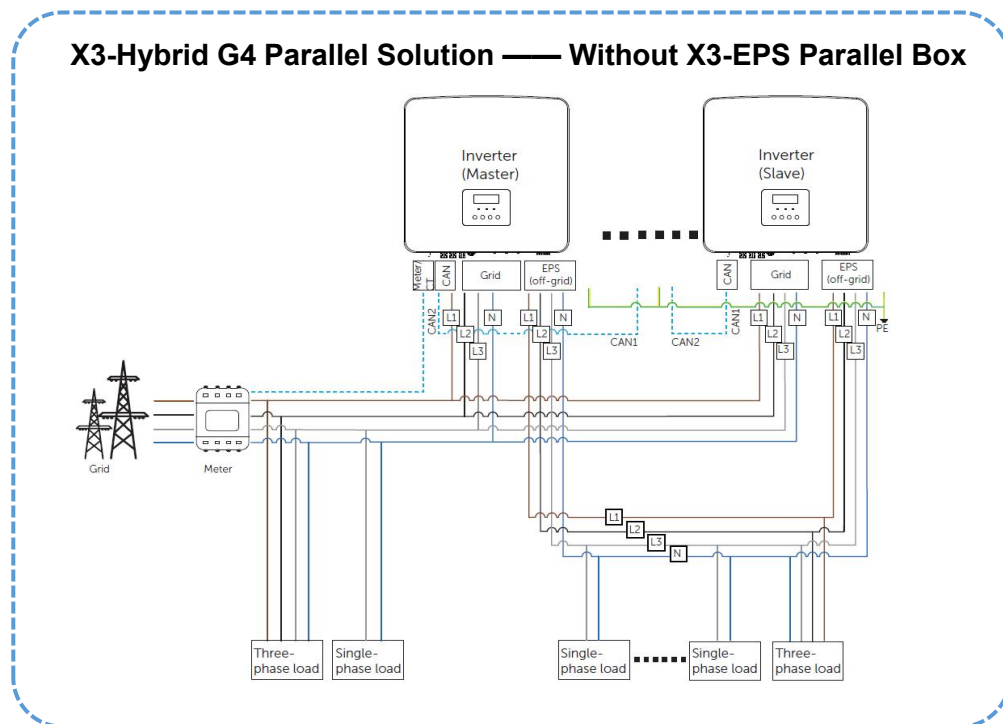
Cable Connection

It is strongly required that every cable must be connected according to correct line sequence (**R-R, S-S, T-T, N-N**), otherwise any small misoperation may cause the system running failed. Incorrect line sequence will damage the inverter. To avoid the damage, the default "Disable" has been set to **"Enable"** in **"External ATS"** under **"Advance Settings"**.

1. The system X3-EPS Parallel Box G2 (Internal ATS)

Please connect the cables strictly according to diagram below.

Up to 3 inverters can be connected in this solution.

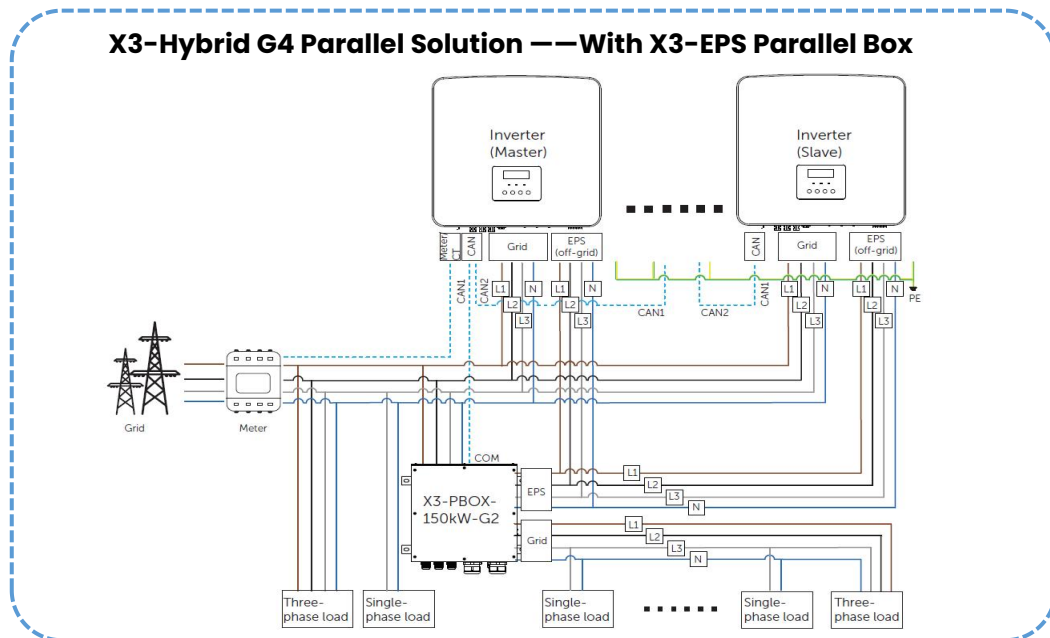


2. The system with X3-EPS Parallel Box G2 (External ATS)

Please connect the cables strictly according to Diagram below.

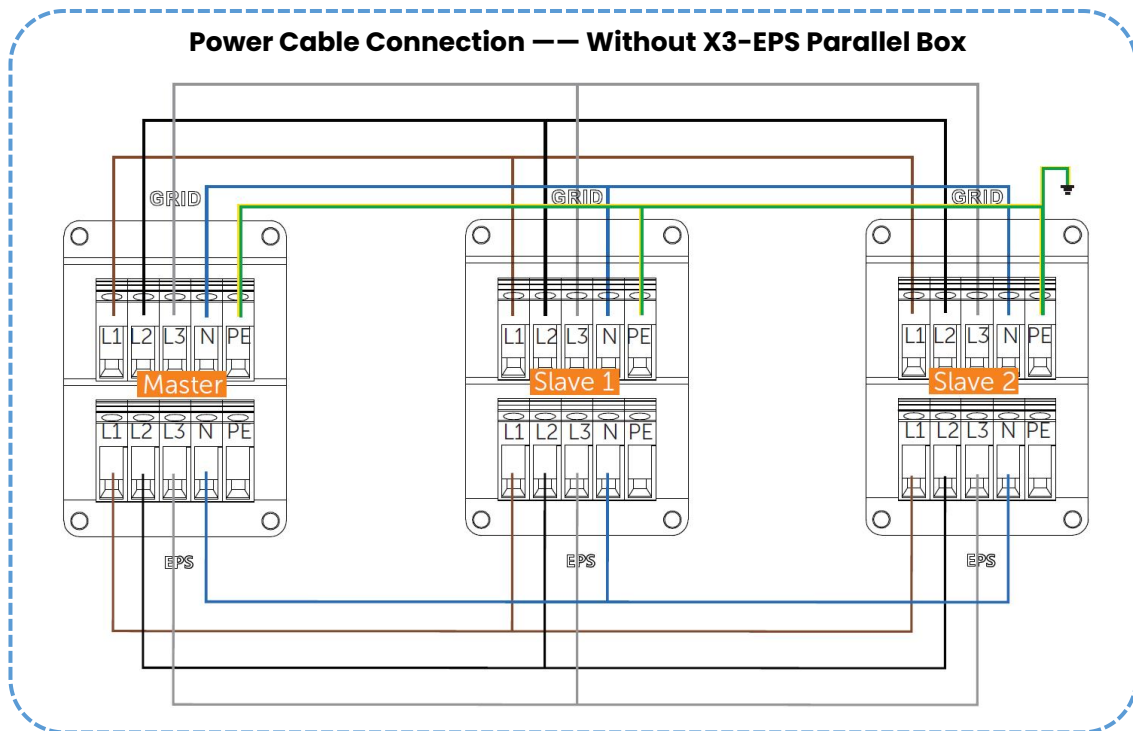
Up to **10 inverters/150kW** can be connected with X3-EPS PBOX-150kW-G2;

Up to **6 inverters/60kW** can be connected with X3-EPS PBOX-60kW-G2.



Power Cable Wiring

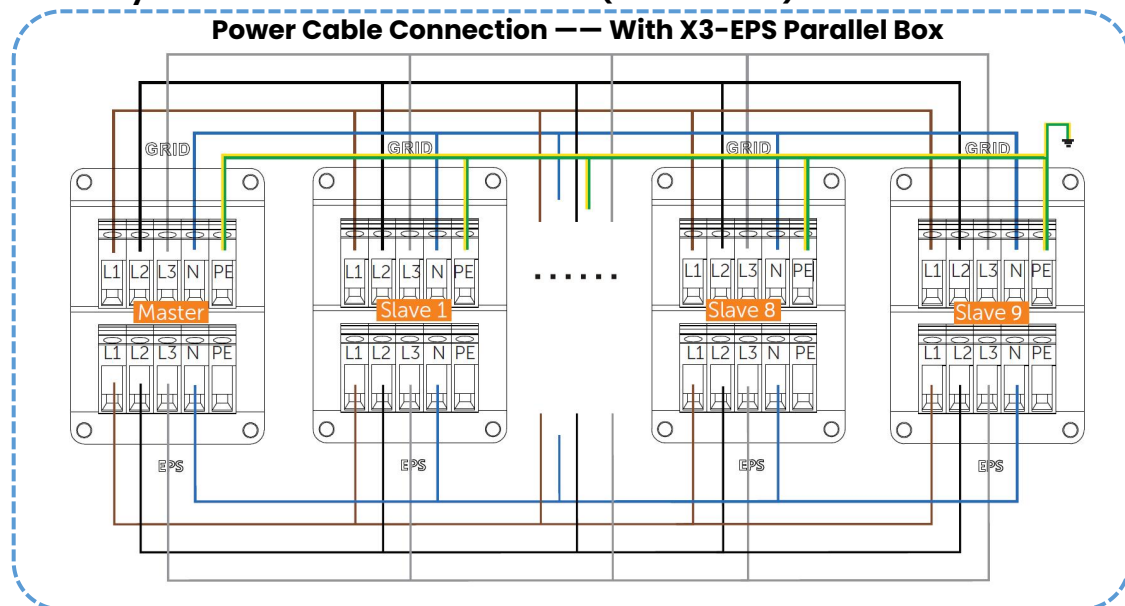
1. The system without X3-EPS Parallel Box G2 (Internal ATS)



Note:

- The system requires five-core copper cable for grid connection and four-core copper cable for EPS connection between master and slave inverter connection.
- Master and Slave inverter's terminal should be connected correspondingly: L1 to L1, L2 to L2, L3 to L3 and N to N. (For both Grid ports and EPS ports connection)
- All PE cable connects to the E-BAR nearby

2. The system with X3-EPS Parallel Box G2 (External ATS)

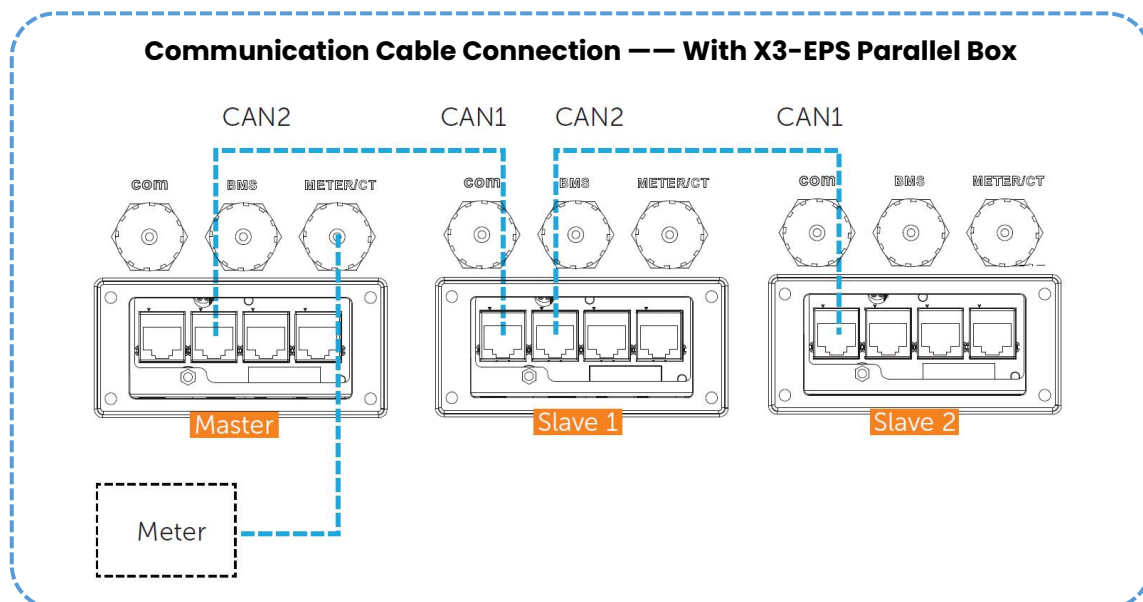


Note:

- The system requires five-core copper cable for grid connection and four-core copper cable for EPS connection between master and slave inverter connection.
- Master and Slave inverter's terminal should be connected correspondingly: L1 to L1, L2 to L2, L3 to L3 and N to N. (For both Grid ports and EPS ports connection)
- All PE cable connects to the E-BAR nearby
- For X3-EPS PBOX-60kW-G2, it supports up to 6 inverters but the total system power should be below than 60kW. For example, for X3 Hybrid G4 15kW, only 4 inverters can use in this system. For X3-EPS PBOX-150kW-G2, it supports up to 10 inverters.

Communication Cable Wiring

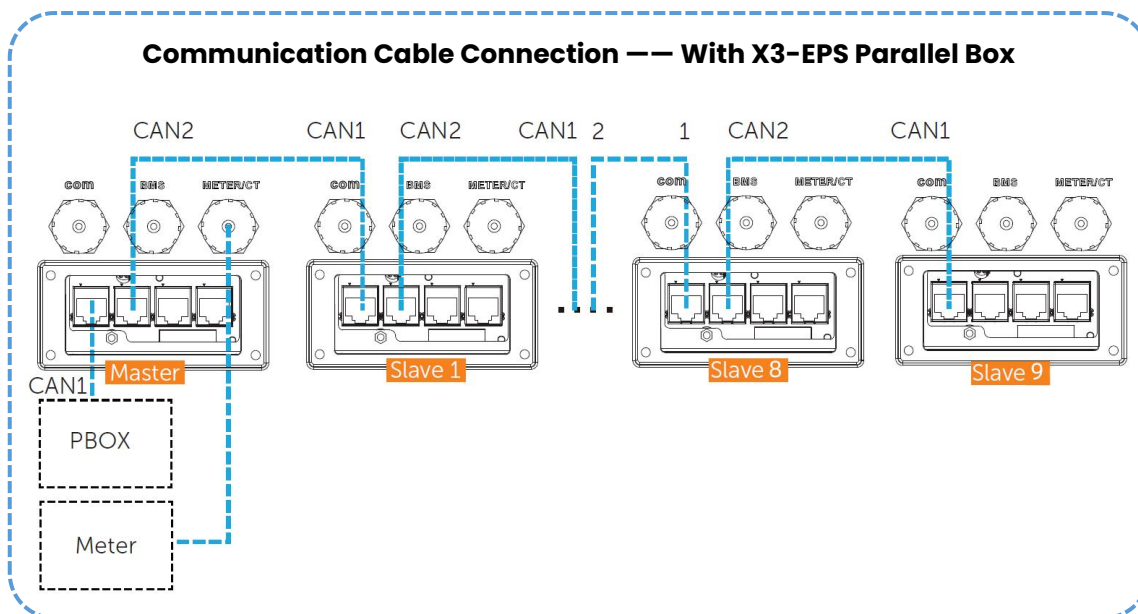
1. The system without X3-EPS Parallel Box G2 (Internal ATS)



Note:

- Use standard network cables for connection between Master inverter and Slave inverter.
- Master **CAN2** connects to Slave 1 **CAN1**.
- Slave 1 **CAN2** connects to Slave 2 **CAN1**; Other inverters are connected in such way.
- Meter connects to **Meter/CT** terminal of the Master inverter.

2. The system with X3-EPS Parallel Box G2 (External ATS)

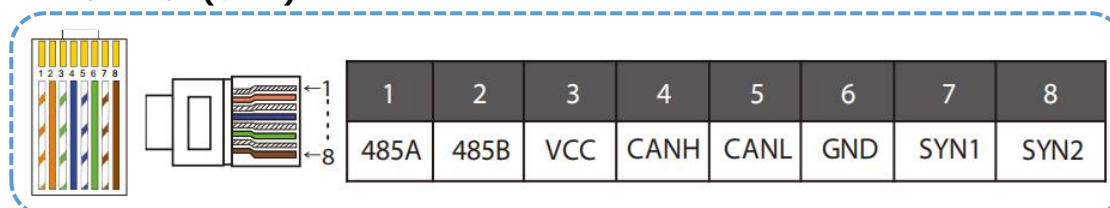


Note:

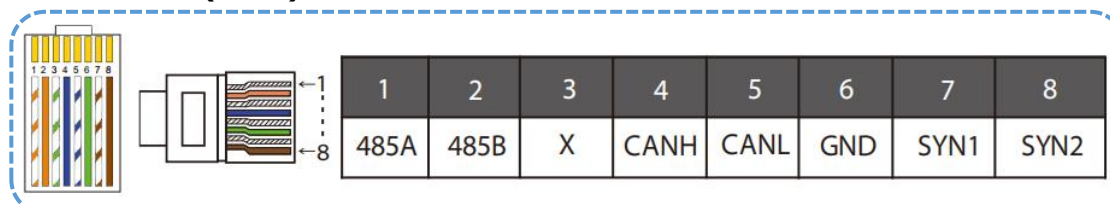
- Use standard network cables for connection between Master inverter and Slave inverter, and master inverter and X3-EPS Parallel Box respectively.
- Master **CAN1** connects to the **COM** terminal of X3-PBOX device.
- Master **CAN2** connects to Slave 1 **CAN1**.
- Slave 1 **CAN2** connects to Slave 2 **CAN1**; Other inverters are connected in such way.
- Meter connects to **Meter/CT** terminal of the Master inverter.

3. X3 Hybrid G4 Pin Definition

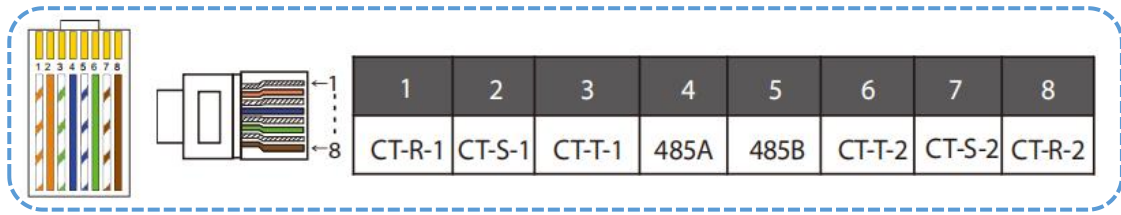
PIN Definition (CAN1)



PIN Definition (CAN2)



PIN Definition (Meter/CT)



Setting & Commission

1. Parallel Setting

Setting path: *Status*

Main → Settings → Advance Setting → Parallel Setting → Parallel Setting.



There are 2 settings chosen:

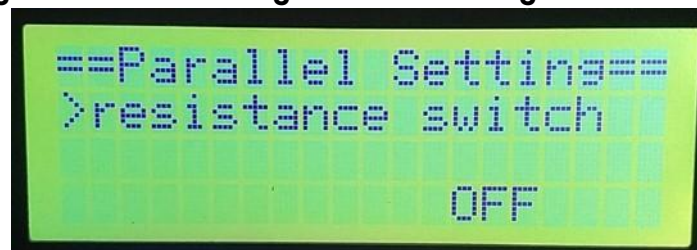
Free

Master

Select **Master** on the **master inverter** (connected to the meter/CT directly), and other inverters will be slave inverters automatically.

Setting path: *Resistance Switch*

Main → Settings → Advance Setting → Parallel Setting → Resistance Switch



There are 2 settings chosen:

OFF

ON

Select **ON** on the **master inverter** and the **last slave inverter** in the parallel system.

Note:

If a slave inverter is set to **Free** mode but not disconnect the network cable, this inverter will return to **Slave** mode automatically.

If a slave inverter is disconnected with master inverter but not be set to **Free** mode, this slave inverter will stop working and prompt **Parallel Fault**.

2. Meter/CT Setting

Setting path:

Main → Settings → Advance Setting → Meter/CT Settings

As for **Internal ATS** solution, 200A CT (309301004800) is optional.

Please select **200** under **Meter/CT Setting → CT Type**.



As for **External ATS** solution with more inverters, a meter with CT is recommended.

Compatibility

Hardware version:

X3-Hybrid/Fit G4.1

X3-Hybrid/Fit G4.3

X3-Hybrid/Fit G4.2

X3-Hybrid/Fit G4.4

Software version:

ARM≥V1.18

DSP≥V1.19